E-Commerce Data Analytics Internship

Business Insights Report

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1. Objective

This report analyzes 51,256 cleaned e-commerce order lines from 2011 to 2014 to find what drives sales and profit, and where the business loses money. The work was done in three stages: data cleaning and pivot analysis in Excel, SQL analysis in MySQL, and an interactive dashboard in Power BI.

2. Data cleaning summary

- Removed 54 exact duplicate rows.
- Filled 79 missing customer names with "Unknown".
- Removed 14 rows with missing profit and 1 row with sales of 0.
- Converted sales from text to numbers (values such as "1,275" contained a comma).
- Standardized the mixed date formats and trimmed extra spaces.
- Kept 12,537 negative-profit rows, because they are real losses caused by discounts.

3. Key performance indicators

KPI	Value	What it means
Total Sales	12,632,373	Revenue from 51,256 cleaned order lines, 2011 to 2014
Total Profit	1,467,499	About 11.6% of sales
Total Orders	25,026	Unique orders (distinct count of order ID)
Average Discount	14.3%	Typical discount across all order lines

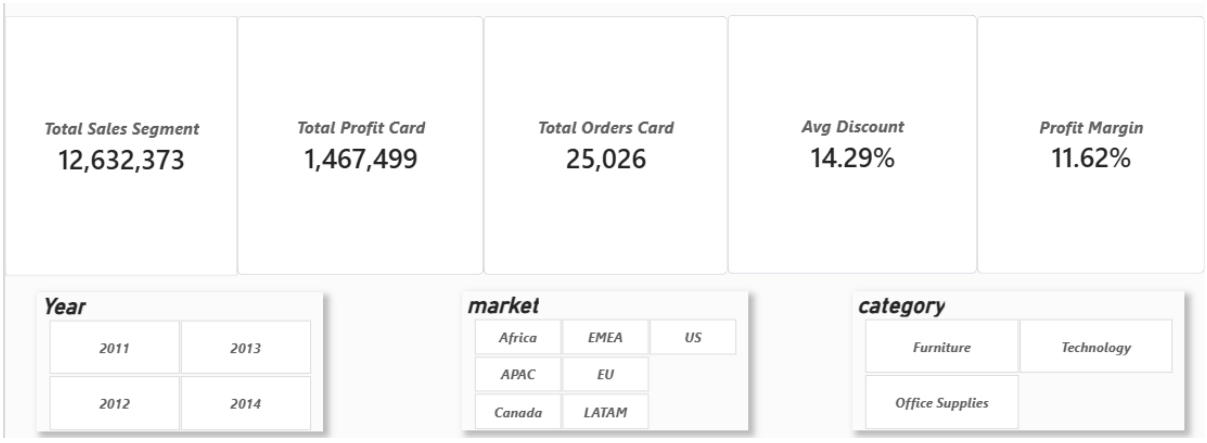


Figure 1: KPI cards and slicers on the Executive Overview page.

4. Sales analysis

Sales grew from 2,256,720 in 2011 to 4,293,920 in 2014. The profit margin stayed near 11.6%, so growth came from selling more rather than earning more on each sale.

Market	Sales	% of sales	Profit margin
APAC	3,581,361	28.4%	12.2%
EU	2,936,867	23.2%	12.7%
US	2,295,756	18.2%	12.5%
LATAM	2,163,876	17.1%	10.2%
EMEA	804,920	6.4%	5.4%
Africa	783,261	6.2%	11.3%
Canada	66,332	0.5%	26.8%

Category	Sales	Profit	Margin
Technology	4,741,813	663,200	14.0%
Furniture	4,105,180	286,125	7.0%
Office Supplies	3,785,380	518,174	13.7%

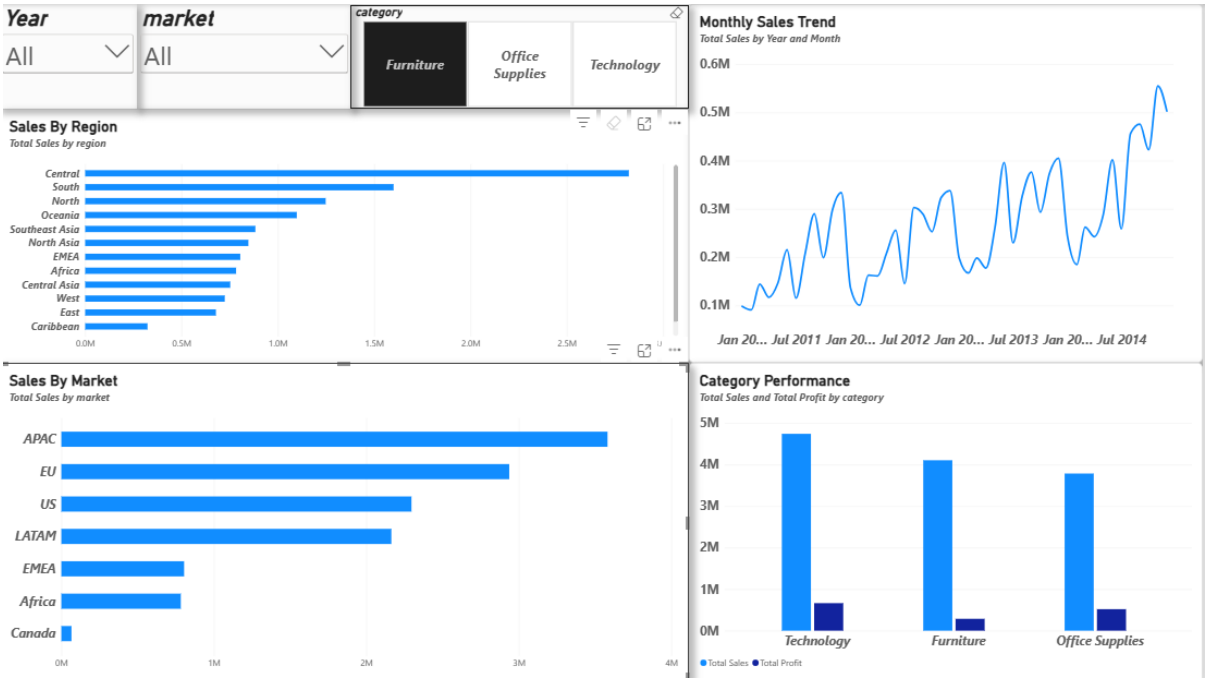


Figure 2: Sales by region, sales by market, monthly sales trend and category performance.

5. Products and customers

Top 5 products by sales	Sales
Apple Smart Phone, Full Size	86,936
Cisco Smart Phone, Full Size	76,441
Motorola Smart Phone, Full Size	73,159
Nokia Smart Phone, Full Size	71,904
Canon imageCLASS 2200 Advanced Copier	61,600

Top 5 loss-making products	Total profit
Cubify CubeX 3D Printer Double Head Print	-8,880
Lexmark MX611dhe Monochrome Laser Printer	-4,590
Motorola Smart Phone, Cordless	-4,447
Cubify CubeX 3D Printer Triple Head Print	-3,840
Bevis Round Table, Adjustable Height	-3,650

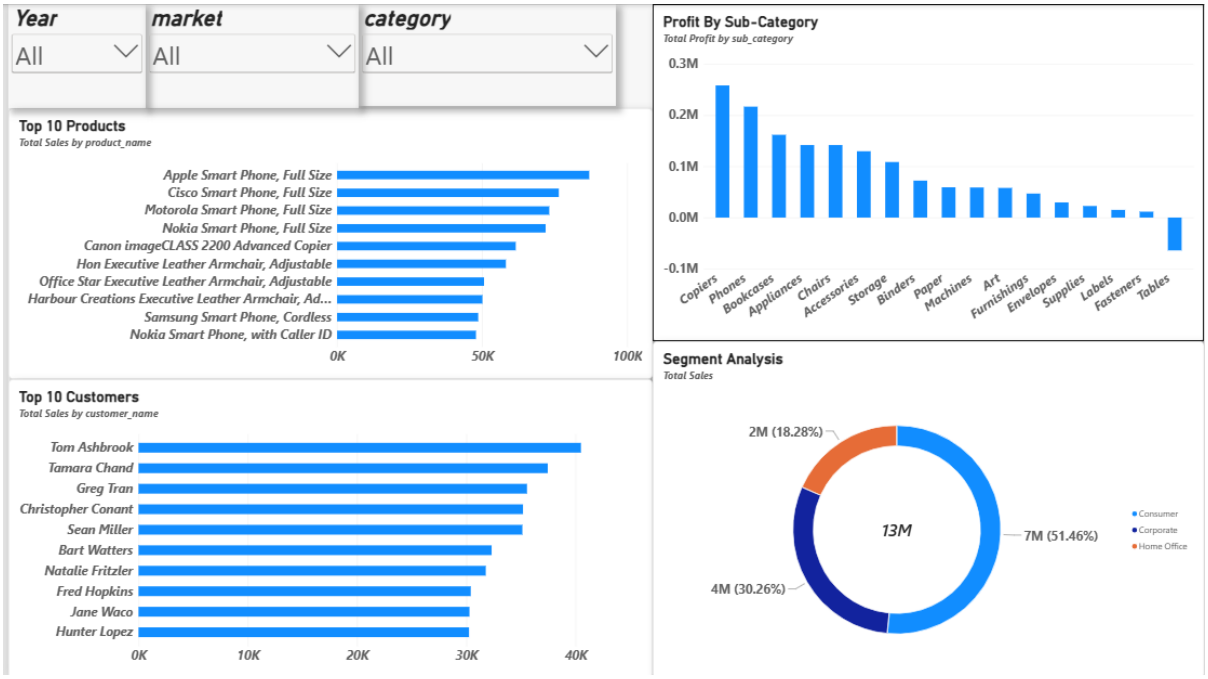


Figure 3: Top products, top customers, profit by sub-category and segment analysis.

6. Impact of discounts

Discount band	Order lines	Sales	Profit	Margin
No discount	28,993	6,988,099	1,770,023	25.3%
0-10%	4,674	1,959,448	339,156	17.3%
10-30%	7,234	2,137,463	151,554	7.1%
30-50%	6,188	1,175,995	-380,922	-32.4%
Above 50%	4,167	371,368	-412,313	-111%

7. Key insights

- Discounts above 30% are the biggest profit problem. Order lines discounted more than 30% lost about 793,235 in total, while orders with no discount earned a 25.3% margin.
- Technology earns about 45% of all profit at a 14.0% margin. Furniture has the lowest margin at 7.0%.
- The Tables sub-category lost 64,113 on sales of 756,031. It is the only loss-making sub-category.
- APAC is the largest market (28.4% of sales). EMEA has the weakest margin at 5.4%, and Canada has the best margin but only 0.5% of sales.

- Turkey (-98,430) and Nigeria (-80,721) lose money overall.
- Standard Class is the most used shipping mode, with 30,755 order lines (60%).
- Sales peak in November and December, which together bring about a quarter of annual sales.

8. Recommendations

- Limit discounts to about 30% to stop the largest losses.
- Review the pricing and discounts on Tables and similar low-margin products.
- Focus marketing on Technology and Office Supplies, which earn the best margins.
- Investigate the loss-making countries before spending more on them.
- Prepare stock and offers ahead of November and December.

9. Conclusion

The data shows a growing business whose profit is held back mainly by heavy discounting and a few weak products and countries. Controlling discounts and focusing on high-margin categories would raise the 11.6% overall margin without needing more sales. The Power BI dashboard lets managers filter by year, market and category to track these areas.